

# Supplementary Material: Cross-Age Face Verification from Mined Same-Person Social-Media Pairs

Danil F. Volkhin and Ivan S. Kopylov

This supplement collects material that supports, but need not occupy, the main paper: (SI) an explicit claims-vs-evidence map; (SII) a consolidated audit of the automatic (LLM / clustering / dedup) supervision components plus a small manual validation (100 posts), with a large multi-annotator audit left beyond this study’s resources; and (SIII) a model / data card. Section and table references of the form “Sec. V” or “Table X” point to the main paper.

## I. CLAIMS VS. EVIDENCE

Table I maps every substantive claim to its evidence and its scope, separating what is *established within the attribution protocol* from what is *not established*. The intent is to make the bounded reading explicit and to forestall over-generalization.

## II. CONSOLIDATED AUDIT OF THE AUTOMATIC SUPERVISION

The supervision is *naturally* (not human-) supervised, so its automatic components are potential hidden label noise. Table II consolidates the cross-checks reported across the main paper into a single view, in the structure a reviewer would expect, and states explicitly which rows are *automatic cross-checks* (available at corpus scale) and which are checked against *human ground truth*. We additionally ran a *small* manual validation (one annotator; 100 posts and sampled pairs/merges) that broadly confirms the automatic pipeline (Table II, lower block); a large multi-annotator audit with inter-annotator  $\kappa$  remains labor-intensive and beyond this study’s resources. We do not treat the LLM-vs-regex agreement as accuracy against a human reference; the “Reference” column makes the distinction.

The automatic rows are reproducible from the released code and aggregates. The manual-validation rows are a *small* single-annotator spot-check (no inter-annotator  $\kappa$ ): they indicate residual error at the few-percent level and corroborate, but do not certify, the automatic pipeline. A large stratified multi-annotator audit (post  $\rightarrow$  group  $\rightarrow$  cluster  $\rightarrow$  pair, with a confusion matrix and  $\kappa$ ) remains labor-intensive and beyond this study’s resources, and the residual noise is acknowledged as a limitation.

## III. MODEL / DATA CARD

**Model.** A deliberately weak trainable face recognizer (FaceNet, Inception-ResNet-v1, casia-webface init) fine-tuned with a pair-contrastive margin objective on mined same-person

pairs; this is an *attribution probe*, not a deployable recognizer.

**Intended use.** Research into whether naturally occurring longitudinal social-media pairs supply useful cross-age identity supervision. Permitted applied analogues are limited to rights-cleared, consent-based, human-in-the-loop settings (Sec. VIII).

**Out of scope / forbidden.** Mass identification, surveillance, de-anonymization, automated decisions, any processing of minors’ data without a legal basis, and use as a general-purpose verifier (low-FAR easy-benchmark performance degrades).

**Training data.** Public multi-photo “then/now” posts (VK communities; an independent Reddit board for the cross-platform test). Supervision is manual-label-free; ages and group integrity come from an LLM caption parser; persons from embedding clustering. Apparent-attribute skew: 76.7% apparent-female. Raw images are *not* released.

**Evaluation data.** External: FG-NET (large-gap  $\geq 25$  y subset), AgeDB-30, CALFW, LFW. Internal: a curated cross-age test (5,107 positives with controlled negatives). Cross-platform: 1,575 Reddit positive pairs.

**Metrics.** ROC-AUC (primary: FG-NET large-gap), EER, TAR@FAR, FMR/FNMR by stratum, bootstrap and identity-level bootstrap CIs, DeLong and paired-permutation tests.

**Ethical considerations.** Biometric special-category data; no individual consent; minors present in the child $\leftrightarrow$ adult regime and withheld from release; weights and embeddings are biometric templates released only under controlled access. Formal ethics approval is pending (Sec. VIII).

**Caveats.** The gain is targeted to the large-gap regime and to weak/medium backbones; external validity beyond the then/now format is not established; the automatic supervision carries residual, not-yet-human-audited noise.

TABLE I

CLAIMS VS. EVIDENCE. “EST.” = ESTABLISHED WITHIN THE WEAK/MEDIUM-BACKBONE ATTRIBUTION PROTOCOL OF THE MAIN PAPER; “BOUNDED” = HOLDS ONLY UNDER STATED CONDITIONS; “OPEN” = NOT ESTABLISHED / OUT OF SCOPE.

| Claim                                                                          | Evidence (main paper)                                                                                                                                                                                 | Status               |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Mined “then/now” pairs improve large-gap ( $\geq 25$ y) cross-age verification | FG-NET large-gap ROC-AUC 0.736 $\rightarrow$ 0.848; DeLong $p=1.4 \times 10^{-20}$ , $\Delta$ AUC 95% CI [+0.088, +0.135]; internal 25+ $p=1.7 \times 10^{-34}$ (Sec. V-A)                            | Est.                 |
| The gain is <i>targeted</i> , not a uniform verification gain                  | AgeDB-30 $-0.008$ ( $p=4 \times 10^{-6}$ ), CALFW null ( $p=0.95$ ), LFW TAR@FAR=0.1% $0.858 \rightarrow 0.575$ (Sec. V-A, Sec. VI)                                                                   | Est.                 |
| The same-person pairing, not the caption age anchor, drives the gain           | Ablation: explicit age anchors add only +0.009 on the 25+ bucket (Sec. V-A)                                                                                                                           | Est.                 |
| Real mined pairs outperform synthetic aging                                    | +real 0.848 vs. FRAN 0.727 and gap-matched FRAN-gm 0.759 (+0.089); native-resolution control (Sec. V-B)                                                                                               | Est.                 |
| The effect reproduces across objective / architecture / source                 | 3 loss families, 2 architectures, 2 communities; ArcFace 0.851 $\approx$ contrastive 0.848; sub-center ArcFace 0.851 (Sec. V-C/E)                                                                     | Est.                 |
| A noise-robust objective adds nothing on curated pairs                         | Sub-center ArcFace ( $K=3$ ) $\equiv$ vanilla ArcFace on FG-NET large-gap (Sec. V-E)                                                                                                                  | Est.                 |
| The mechanism is removal of an age shortcut                                    | Under age-matched negatives frozen falls 0.640 $\rightarrow$ 0.517 ( $\approx$ chance); pairs restore 0.838 (Sec. V-D)                                                                                | Est.                 |
| The supervision is data-efficient                                              | $\approx 96\%$ of the gain at 10% of identities (Sec. V-C)                                                                                                                                            | Est.                 |
| No apparent-demographic stratum degrades; errors fall everywhere               | Stratified AUC (all CIs $> 0$ ); FMR/FNMR at 1% FMR: FNMR down in every stratum (Sec. V-G)                                                                                                            | Est.                 |
| About half of the raw “positives” were near-duplicates                         | 65,897 $\rightarrow$ 31,865 ( $-52\%$ ), stable over cosine 0.93–0.97 (Sec. III, Table XI)                                                                                                            | Est.                 |
| The gain transfers beyond the originating platform, in <i>both</i> directions  | VK $\rightarrow$ Reddit overall ROC-AUC 0.718 $\rightarrow$ 0.749; Reddit $\rightarrow$ VK FG-NET large-gap 0.736 $\rightarrow$ 0.783 (+0.047 from a 220-person source); small, noisy sets (Sec. V-H) | Bounded              |
| The contribution is specific to weak/medium backbones                          | Strong saturated backbones show null/negative gains (Sec. V-C)                                                                                                                                        | Bounded (limitation) |
| Platform-agnostic / cross-cultural generalization                              | one platform family, then/now format; single non-VK test                                                                                                                                              | Open                 |
| Drop-in upgrade to a general-purpose verifier                                  | low-FAR easy-benchmark operating points degrade                                                                                                                                                       | Open (excluded)      |
| Human-grade label-noise guarantee                                              | automatic cross-checks + a small manual validation (100 posts; SII); no multi-annotator $\kappa$                                                                                                      | Bounded              |
| Full SOTA-pipeline parity                                                      | only the shared margin objective is trained, not method-specific add-ons                                                                                                                              | Open (by design)     |
| Legal/ethical clearance for biometric processing                               | research basis stated; formal approval pending (Sec. VIII)                                                                                                                                            | Open (author/IRB)    |

TABLE II

CONSOLIDATED AUDIT OF THE AUTOMATIC SUPERVISION. “REFERENCE” STATES WHAT THE METRIC IS MEASURED AGAINST: AN AUTOMATIC CROSS-CHECK (UPPER BLOCK, CORPUS SCALE) OR HUMAN GROUND TRUTH FROM A SMALL MANUAL VALIDATION (LOWER BLOCK; 100 POSTS + SAMPLED PAIRS/MERGES, ONE ANNOTATOR).  $n$  IS THE AUDITED POPULATION. A LARGE MULTI-ANNOTATOR AUDIT WITH  $\kappa$  REMAINS BEYOND RESOURCES.

| Audit target                                                                     | $n$                  | Reference             | Result                                                                                                 |
|----------------------------------------------------------------------------------|----------------------|-----------------------|--------------------------------------------------------------------------------------------------------|
| Age extraction (LLM vs. regex)                                                   | 33,697 captions      | automatic cross-check | 72.8% agree; on the 27.2% disagreements the LLM is favored (+10.8% filled, $-2.5\%$ missed); Table III |
| Group integrity (LLM classifier)                                                 | 33,697 posts         | automatic (LLM)       | 92.8% single-person, 6.0% multi-person, 1.0% unknown, 0.2% meme; Table IV                              |
| Multi-person noise detector (within-person gender consistency)                   | multi-photo groups   | automatic proxy       | 2,919 groups with consistency $< 0.6$ flagged as probable multi-person                                 |
| Near-duplicate inflation (within-person dedup)                                   | 65,897 raw positives | automatic (embedding) | $-52\%$ ( $\rightarrow$ 31,865); stable for cosine 0.93–0.97; Table XI                                 |
| Person-clustering recurrence                                                     | 27,487 post-groups   | automatic (embedding) | 21,848 persons ( $\approx 20\%$ cross-post recurrence); conservative cosine 0.85 merge                 |
| <b>Small manual validation</b> (one annotator; 100 posts + sampled pairs/merges) |                      |                       |                                                                                                        |
| Age extraction vs. human                                                         | 100 captions         | human gold            | exact-age 89%, within $\pm 1$ y 95% (regex 71% exact); MAE 0.5 y                                       |
| Group integrity (single vs. multi)                                               | 100 posts            | human gold            | LLM precision 0.96, recall 0.93                                                                        |
| Same-person positive pairs                                                       | 200 pairs            | human gold            | precision 0.96 (8/200 different-person)                                                                |
| Cluster merges (cosine $\geq 0.85$ )                                             | 100 merges           | human gold            | 95% correct                                                                                            |
| Near-duplicate removals                                                          | 100 removed          | human gold            | 93% genuine duplicates                                                                                 |